Palindromic Proper-Divisor Residues at Every Modulus

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

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Abstract

For every $n \geq 1$, we construct an integer with exactly $2n+2$ positive divisors whose $2n+1$ proper-divisor residues modulo n form a palindrome. This proves the existence conjecture in OEIS A393833. The construction needs no prime-distribution theorem: a prime divisor of a suitable cyclotomic value suffices. An explicit witness bound makes every minimum a finite arithmetic minimization over ordered multiplicative partitions of $2n+2$ and prime tuples. An exact factorization search implements that formula.

1 An elementary witness for every index

For $K > 1$ with ordered divisors $1 = d_1 < d_2 < \cdots < d_D = K$, let its proper-divisor residue list be $(d_1 \bmod n, \ldots, d_{D-1} \bmod n)$. OEIS A393833 asks for the least such K with $D = 2n+2$ and a palindromic list, and conjectures that it exists at every positive index.

Lemma 1. *For every $n \geq 2$ there is an odd prime $p \equiv 1 \pmod{n}$. More explicitly, every prime divisor of $\Phi_n(2n)$ has these properties, where Φ_n denotes the nth cyclotomic polynomial.*

Proof. For $n > 1$, the monic integer polynomial Φ_n has constant term one. Its roots have absolute value one, so

$$\Phi_n(2n) = \prod_{\zeta \text{ primitive } n\text{th root}} (2n - \zeta) > 1;$$

indeed its absolute value is at least $(2n-1)^{\varphi(n)} > 1$, and it is positive. Choose a prime p dividing this value. No prime divisor of $2n$ divides the value, by its constant term. Thus p is odd and $p \nmid n$. Over $\mathbf{F}_p$, the polynomial $X^n - 1$ is square-free, since its derivative is nX^{n-1}. The factorization $X^n - 1 = \prod_{d|n} \Phi_d(X)$ therefore has pairwise coprime factors modulo p. The residue of $2n$ is a root of Φ_n, so its multiplicative order is exactly n: an order $d < n$ would also make it a root of a factor of $X^d - 1$, contradicting that coprimality. The order divides $p-1$, proving $p \equiv 1 \pmod{n}$. □

Theorem 1. *Let $n \geq 1$, and let p be any odd prime congruent to one modulo n. Then*

$$K = 2p^n$$

has exactly $2n+2$ divisors and its proper-divisor residues form a palindrome. Thus A393833 has a positive value at every index.

Proof. Since $p > 2$, its ordered divisors are

$$1, 2, p, 2p, p^2, 2p^2, \ldots, p^n, 2p^n.$$

There are $2(n+1)$ of them. After deleting the final divisor, their residues alternate $1, 2, 1, 2, \ldots, 1$ modulo n, a list of odd length and hence a palindrome. For $n = 1$ use $p = 3$; for $n \geq 2$, Lemma 1 supplies a suitable prime. □

Define p_n to be the least odd prime congruent to one modulo n and $B_n = 2p_n^n$. This is an explicit finite arithmetic construction: $p_1 = 3$, and for $n \geq 2$ one need test primes only up to $\Phi_n(2n)$ by Lemma 1. Consequently

$$a(n) \leq B_n \leq 2\Phi_n(2n)^n \qquad (n \geq 2). \tag{1}$$

2 An exact finite minimum formula

Put $D = 2n+2$. An ordered multiplicative partition of D is a tuple $(f_1, \dots, f_t)$ with $f_i \geq 2$ and $\prod_i f_i = D$; order is retained. There are finitely many, with $1 \leq t \leq \lfloor \log_2 D \rfloor$. For such a tuple and primes $q_1 < \cdots < q_t$, form

$$M(\mathbf{q}, \mathbf{f}) = \prod_{i=1}^{t} q_i^{f_i - 1}, \qquad \mathcal{D}(\mathbf{q}, \mathbf{f}) = \operatorname{sort}\left\{ \prod_{i=1}^{t} q_i^{j_i} : 0 \leq j_i < f_i \right\}.$$

The latter list has exactly D elements; write them as $c_1 < \cdots < c_D$. Define the finite candidate set

$$\mathcal{C}_n = \big\{ M(\mathbf{q}, \mathbf{f}) : \prod_i f_i = D, \quad q_1 < \cdots < q_t \leq B_n, \\ M(\mathbf{q}, \mathbf{f}) \leq B_n, \quad c_i \equiv c_{D-i} \pmod{n} \ (1 \leq i < D) \big\}.$$

Theorem 2. *For every $n \geq 1$, the exact least value is*

$$\boxed{a(n) = \min \mathcal{C}_n.}$$

In particular the zero escape value never occurs. Any independently verified witness can replace B_n throughout the formula.

Proof. Theorem 1 places B_n in $\mathcal{C}_n$, using the ordered bases $(2, p_n)$ and factors $(2, n+1)$. Every member has exactly D divisors, and the displayed congruences are exactly the palindrome condition on the proper-divisor list. Conversely a least admissible integer is at most B_n. Its unique prime factorization has distinct increasing bases q_i and positive exponents $f_i - 1$ with $\prod_i f_i = D$; all its bases are at most B_n. Thus it belongs to the candidate set. □

This formula is a finite arithmetic minimization, not a claim of an elementary closed expression without a minimum. The substantive new global input is the explicit witness and bound, which removes the unbounded existence search in the original definition.

3 Certified pruning and implementation

Two elementary facts greatly shorten the factor-pattern search. First, any admissible K satisfies

$$K/q_1 \equiv 1 \pmod{n}, \tag{2}$$

because its largest proper divisor must have the same residue as its first proper divisor, one. Hence any partial factorization V whose smallest prime is q_1 can be discarded as soon as $\gcd(V/q_1, n) > 1$; multiplication cannot repair this obstruction.

Second, define $E(R) = \sum_{\ell^a \| R} a(\ell - 1)$ for prime ℓ. For any multiplicative partition $R = \prod f_i$,

$$\sum_i (f_i - 1) \geq E(R). \tag{3}$$

Indeed $(uv - 1) \geq (u-1) + (v-1)$ for integers $u, v \geq 2$; splitting every factor into primes proves the inequality. At a partial factorization V, choose the next exponent $e = f - 1$, where $f \mid R$, $f > 1$, and the next prime q. Every completion then has value at least $Vq^{e+E(R/f)}$. With current verified upper bound H, one need enumerate only

$$q \leq \left\lfloor \big((H-1)/V\big)^{1/(e+E(R/f))} \right\rfloor.$$

All roots and powers in the evaluator use integer arithmetic. It enumerates every divisor $f > 1$ of the remaining divisor count, so it retains all ordered exponent patterns, including those whose exponents increase with their bases. It applies (2) before the final sorted-divisor palindrome check. Validated published terms may initialize H; they serve only as witnesses, and every smaller candidate is still excluded by the complete search.

The independent verification uses a divisor sieve through $K = 200000$, compares all minima it encounters with the factor-pattern evaluator, checks all 13 terms stated in the official entry and its comments, and checks the explicit witnesses through $n = 100$. A b-file gives exact computed minima through $n = 12$, including an index absent from the entry's displayed data. These finite checks supplement the proofs; the proofs establish existence and the finite formula at every positive index.

References

[1] The OEIS Foundation Inc., A393833, accessed October 26, 2026.